Kavya Varma

Embedded Firmware Engineer

RESUME_1202

Location: Rajkot, Gujarat, India | Email: kavya.varma.resume_1202@example.com | Phone: +91-00000-00000 (placeholder)
Total Experience: 1 year

PROFESSIONAL SUMMARY

Embedded Firmware Engineer with 1 year of experience in embedded firmware, using C, C++, ARM Cortex-M, FreeRTOS. Experienced in real-time systems, hardware debugging and measurable delivery outcomes.

TECHNICAL SKILLS

Skills: I2C, CAN, JTAG, ARM Cortex-M, Git, FreeRTOS, STM32, UART, Firmware Debugging, SPI, C, C++

Methods: Real-time systems, Hardware debugging, Protocol validation

PROFESSIONAL EXPERIENCE

Embedded Firmware Engineer | NovaPeak Engineering | Jul 2025 - Present

- Lowered field-reset incidents by 27% with watchdog recovery and diagnostics
- Improved battery runtime by 22% with peripheral and sleep-state optimization
- Reduced embedded task jitter by 41% using interrupt priority and queue tuning

PROJECTS

Motor Control Interface | UART, JTAG, ARM Cortex-M, C++

- Implemented CAN message handling, fault logging and deterministic state transitions.
- Documented acceptance criteria and evaluated the design using hardware debugging and protocol validation.

Low-Power Metering Device | Git, CAN, SPI, FreeRTOS

- Optimized interrupt-driven acquisition and sleep cycles.
- Documented acceptance criteria and evaluated the design using hardware debugging and protocol validation.

EDUCATION

B.Tech in Electronics and Communication Engineering, Eastgate Institute of Engineering, 2025
Relevant Methods: Real-time systems, Hardware debugging, Protocol validation

CERTIFICATIONS

RTOS Scheduling Training (2026)